

Multiple Choice (10 marks)

1. The Krebs cycle in humans occurs in the
 - (a) peroxisome
 - (b) cytoplasm
 - (c) nucleus
 - (d) intermembrane space
 - (e) outer mitochondrial membrane
2. A rise in intracellular free calcium in the sea urchin oocyte causes the release of proteolytic enzymes which act to prevent polyspermy. The events just described entail the
 - (a) polyspermy reaction
 - (b) enzymatic reaction
 - (c) proteolytic reaction
 - (d) acrosomal reaction
 - (e) oocyte reaction
3. A plant grows in the opposite direction of the gravitational force. This is an example of
 - (a) positive chemotropism
 - (b) positive phototropism
 - (c) negative phototropism
 - (d) negative chemotropism
 - (e) positive hydrotropism
4. Sexual dimorphism is most often a result of
 - (a) parallel selection.
 - (b) sympatric selection.
 - (c) artificial selection.
 - (d) allopatric selection.
 - (e) directional selection.
5. Which of the following adaptations would limit pollination by bees and promote hummingbird pollination?

- (a) Flowers with a narrow opening that only small insects can access
- (b) Pendant (hanging) red-colored flowers
- (c) Modified petals to provide a landing space
- (d) Flowers with a strong sweet fragrance
- (e) Bright blue flowers that open in the early morning

True / False (10 marks)

Answer True or False

- 6. True or False: Does increased nerve length within the treatment volume improve trigeminal neuralgia radiosurgery?
- 7. True or False: Is robotically assisted laparoscopic radical prostatectomy less invasive than retropubic radical prostatectomy?
- 8. True or False: Is gastric electrical stimulation superior to standard pharmacologic therapy in improving GI symptoms, healthcare resources, and long-term health care benefits?
- 9. True or False: Juvenile osteochondritis dissecans: is it a growth disturbance of the secondary physis of the epiphysis?
- 10. True or False: Does binge drinking during early pregnancy increase the risk of psychomotor deficits?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the inheritance patterns of red-green color blindness in a family where a red-green color-blind man marries a woman with normal vision, whose father was color blind. Analyze the expected phenotypic ratios of their offspring.
- 12. Discuss the classification of algae within biological kingdoms, analyzing their characteristics and evolutionary significance in the context of Fungi, Plantae, and Archaea.

End of Paper